MAE 3240 Heat Transfer

Spring 2026

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Acknowledgement:
I referenced the equations on the class slides #2 and #3

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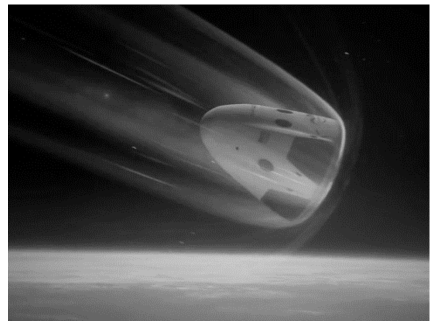
Problem Set #1

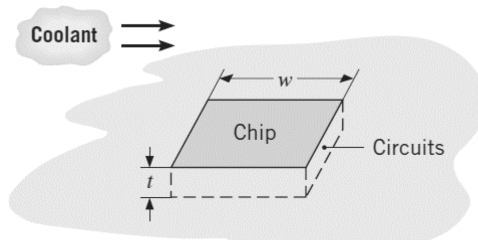
Posted: Monday, Jan 26 **Due: Tuesday, Feb 3, 11:59 pm**

1. (30 pt) Consider a spacecraft traveling in the space *outside* the atmosphere, discuss the dominant heat transfer mode across the spacecraft and briefly explain. After completing the space exploration mission, this spacecraft returns to earth by traveling *through* the atmosphere. Please discuss the dominant heat transfer mode across the spacecraft and briefly explain.



Note: you do not need to follow the recommended format for this problem, since this is a short-answer problem.

2. (30 pt) A square silicon chip (k = 150 W/(m·K)) is of width w = 5 mm on a side and of thickness t = 1 mm. The chip is mounted in a substrate such that its side and back surface are thermally insulated. The front surface is exposed to a coolant with convective flows. The chip dissipates 9 W energy under steady-state operation. Assume that thermal radiation is negligible in this problem. The coolant temperature is T_∞ = 20 °C.



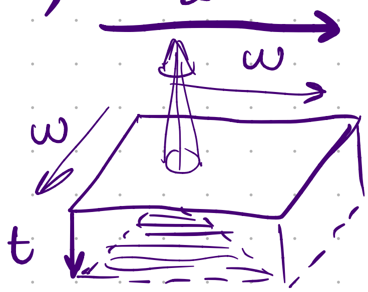
(a) What is the steady-state temperature difference between back and front surfaces of the chip?

(b) If the front surface temperature of the chip cannot be above 80 °C, calculate the minimum heat transfer coefficient required for the coolant flow. Compare your calculation with heat transfer coefficients shown in Table 1.1 of the textbook and discuss what convection processes can be used to cool this chip.

1) heat transfer mode through space would primarily be conduction, as there is a large heat gradient of room temperature inside and temperature of space outside is close to 0. $\Delta T \approx 273K$.
and no fluid medium for convection.

however, during entry, convection takes a lead as the atoms in the atmosphere move relative to the spacecraft, forcing convection over surface.

2) $T_\infty = 20°C$



9W dissipation   no radiation

coolant $T_\infty = 20°C$ $k = 150 \frac{W}{mK}$

$L = 1mm$

Steady state difference?

$$q_x = -kA\frac{dT}{dx} = -kA\frac{T_2 - T_1}{L}$$

$$9 = -150(.005)^2\left(\frac{20 - T_1}{.001}\right)$$

$$-1\left(\frac{9(.001)}{-150(.005)^2} - 20\right) = T_1 = 22.4°C$$

$\boxed{\text{(a) steady state temperature difference of } 2.4°C}$

(b) $q = hA(T_s - T_\infty)$ $T_{smax} = 80$

$T_\infty = 20$

$$\frac{9}{60(.005)^2} = h$$

$\boxed{\text{(b) } 6000 \frac{W}{m^2 K} = h}$ ⇒ according to table 1.1, forced convention with a liquid medium would suit this system best, as it has an h range of $100 - 20,000 \frac{W}{m^2/K}$

3. (40 pt) The emissivity of a roof material is $\varepsilon = 0.18$. The absorptivity of this material for solar irradiation is $\alpha_s = 0.76$.

(a) Consider a day of solar irradiation G_s = 750 W/m^2, ambient temperature T_∞ = 300 K, surrounding temperature T_{surr} = 300 K, and heat transfer coefficient of air flow h = 10 W/(m$^2 \cdot$ K), calculate the temperature of the roof. Assume the bottom surface of the roof is thermally insulated. Assume the roof can be treated as a grey body to the irradiation of surrounding environment. Would the neighboring cat be comfortable walking on the roof constructed by this material?

(b) According to your calculation, discuss the dominant heat transfer mode in (a).

3) emmisivity: $\varepsilon = .18$

absorbivity: $\alpha_s = .76$

a) $G_s = 750\ W/m^2$ $\quad T_\infty = 300K = T_{surr}$

$h = 10\ W/m^2K$

$\downarrow 750 \frac{W}{m^2}$ 300K

$T_\infty \cdot h = 3000 \frac{W}{m^2}$

insulated

absorbed heat -

incident solar = $.76 \cdot G_s$ $\quad q_s = \alpha_s \cdot G_s$

surrounding irrad = .18 surrounding irrad $\quad q_s = .76 \cdot 750$

$q_{solar} = q_{rad} + q_{conv}$

$$.76(750) = \varepsilon A \sigma T_s^4 - \alpha A G + hA(T_s - T_\infty)$$

$G = \sigma T_{sur}^4 \quad \varepsilon = \alpha$ (grey body)

$$\varepsilon A \sigma (T_s^4 - T_{sur}^4) + hA(T_s - T_\infty)$$

$$T_s^4(\varepsilon A \sigma) - (\varepsilon A \sigma T_{sur}^4) + hAT_s - hAT_\infty$$

$$.76(750) + \varepsilon A \sigma T_{sur}^4 + hAT_\infty = T_s^4(\varepsilon A \sigma) + T_s(hA)$$

$$.76(750) + \varepsilon \sigma T_{sur}^4 + hT_\infty = T_s(T_s^3 \varepsilon \sigma + h)$$

$\rightarrow$ Plug in values, solve for T_s

(b) convection is the dominant heat transfer mode, mainly due to the constant σ in radiation, which is in the order of magnitude of 10^{-8} (which compensates for the temperature being order of magnitude 10^{4})